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https://doi.org/10.1038/s41565-018-0120-4
nature nanotechnology

A framework for sustainable nanomaterial selection and design based on performance, hazard, and economic considerations

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Engineered nanomaterials (ENMs) and ENM-enabled products have emerged as potentially high-performance replacements to conventional materials and chemicals. As such, there is an urgent need to incorporate environmental and human health objectives into ENM selection and design processes. Here, an adapted framework based on the Ashby material selection strategy is presented as an enhanced selection and design process, which includes functional performance as well as environmental and human health considerations. The utility of this framework is demonstrated through two case studies, the design and selection of antimicrobial substances and conductive polymers, including ENMs, ENM-enabled products and their alternatives. Further, these case studies consider both the comparative efficacy and impacts at two scales: (i) a broad scale, where chemical/material classes are readily compared for primary decision-making, and (ii) within a chemical/material class, where physicochemical properties are manipulated to tailor the desired performance and environmental impact profile. Development and implementation of this framework can inform decision-making for the implementation of ENMs to facilitate promising applications and prevent unintended consequences.

Engineered nanomaterials (ENMs) are poised to address societal challenges in the fields of medicine, electronics and water treatment[1]. However, environmental impacts associated with ENMs and their manufacture[2,3] could potentially offset the intended benefits. While materials scientists are accustomed to optimizing functional performance and price, minimizing human health or environmental impacts a priori has not been systematically pursued[3]. Thus, ENMs and ENM-enabled products originally intended to solve societal challenges could result in unintended consequences, threatening the long-term commercial viability of ENMs due to real or perceived risks. Furthermore, as the rate of material innovation is rapid and relevant regulations for systematic assessment are lagging[4], health and environmental impact assessments are challenged to keep pace and are therefore conducted on a case-by-case basis.

The significant economic and performance potential for ENM-enabled products is reflected by the United States (US) National Nanotechnology Initiative (NNI) annual investment, totalling approximately US$15 billion from 2007 to 2015 (Fig. 1). Over this same time period, the annual NNI investment in environment, health and safety (EHS) research ranged from US$48.3–120.2 million, never exceeding 8% of the total NNI investment (Fig. 1)[5–13]. This disparity underscores the challenge facing designers who aim to develop sustainable ENMs and ENM-enabled products that maximize functional performance while minimizing adverse environmental or human health impacts. The data necessary to systematically assess safety before commercialization are lacking, which has historically led to two 'fatal' outcomes for emerging technologies: (1) threatened acceptance due to public perception of a risk in the absence of high-quality data, or (2) delayed observation of adverse impacts, leading to 'regrettable substitutions'[14]. The most notable example of the former is the effective elimination of the genetically modified organisms market across much of Europe[15], demonstrating the need for improved design guidelines that support simultaneous consideration of functional performance, price and environmental impact. Such systematic assessments of trade-offs are imperative to inform sustainable design of emerging products, while prospective evaluations are necessary to minimize unforeseen implications associated with material selection and product design decisions.

There have been multiple strategies proposed to address these trade-offs in nanotechnology and nano-enabled products, including multi-criteria decision analysis (MCDA) coupled with life-cycle assessment (LCA)[16,17] and tools such as the LICARA (Life Cycle and Risk Assessment) nanoSCAN[18]. While valuable, these tools can be too qualitative and limited in nuance within each class of nanomaterial to inform design decisions. Additionally, LCAs are time and resource intensive, and result in potentially high levels of uncertainty given limited data available for nanotechnology. Instead, sustainable nanomaterial selection analogous to those used by mechanical and materials engineers, such as the well-established Ashby materials selection framework[19,20], is developed and explored here.

Simplified, Ashby advises a stepwise process to inform material choice by delineating (multi)parametric design requirements and then screening vast material libraries to curate a selection of viable candidates. Material selection charts, which succinctly display materials by class (for example, metals, polymers or composites) versus functional or economic metrics (for example, tensile strength, density, thermal conductivity or price per unit mass)

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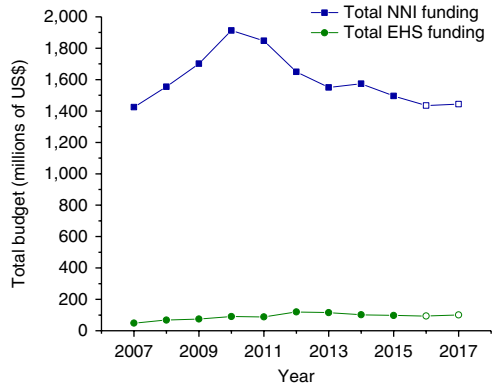


Fig. 1 | Portion of total National Nanotechnology Initiative funding dedicated to environment, health and safety. NNI funding 2007–2015[5–13], including environment, health and safety (EHS) research and estimated or proposed budgets (unfilled data points, 2016–2017).

are central and essential to this process. This is a well-established approach used by both product designers and materials scientists, and its expansion to include nanomaterials has also been previously suggested[1]. However, the systematic evaluation of ENM product performance, and the influence of embedding those ENMs into other carriers (that is, metamaterials), has limited development of ENM-specific selection diagrams. Further, while there have been efforts to use Ashby's material selection framework to assess the environmental implications of some bulk materials[21] and composites[22], understanding the functional performance and environmental impacts of ENMs simultaneously has not been comprehensively pursued. Recently, Ashby expanded metamaterial performance metrics to incorporate environmental life cycle impacts, such as cumulative energy demand (CED), water use and greenhouse gas emissions[21], but notably, there is an opportunity for the inclusion of omitted metrics (for example, toxicity). As such, there is a need to further enhance the framework to include inherent material hazard and environmental or human health implications. The emerging data associated with LCAs, toxicity and functional performance in the novel domain of ENM-enabled products can be harnessed to develop a material selection and design strategy to guide the design of sustainable ENM and ENM-enabled products.

Here, we demonstrate the potential of this strategy via two case studies: (1) selecting between ENMs and bulk chemicals for antimicrobial applications and (2) selecting an ENM to nano-enable a material for high electrical conductivity applications. In addition to the functional performance, design metrics include human health impacts using zebrafish toxicity as a model organism[23,24], cradle-to-gate CED and material price. The approach presented here informs where best to focus limited time and resources in determining the viability of an ENM for a given application. For example, instead of investing in more complicated toxicity assays across multiple trophic levels, or relying on the uncertain nature of life-cycle toxicity assessment, the candidate materials can be curated to those that meet goals set for functionality and sustainability. The success of this approach will rely on a collaboration between materials scientists creating the nanomaterials, environmental scientists evaluating the nanomaterials, and product designers to ensure that (1) robust data are generated relating functional benefits to ENMs and their properties and (2) environmental implications of emerging ENMs can be well understood and communicated, so future iterations of nanomaterials and nano-enabled products can be designed to avoid these issues. In the absence of this concerted effort, nanotechnology may not realize its full potential to have a lasting, positive impact on science and society.

Adapted framework for sustainable ENMs

The potential to support sustainable ENM selection and design through the proposed framework is demonstrated through two case studies. The first highlights the use of ENMs versus bulk chemical substances for antimicrobial activity and the second is applied to ENM selection for nano-enabled polyaniline (PANI) for improved electrical conductivity. These case studies were selected due to real innovation needs driven by regulatory changes, as in the case of recently banned chemical antimicrobials[25], or market demands, as in the case of ENM–PANI composites for flexible supercapacitors[26]. The specific ENMs, chemicals and materials included in each case study were chosen based on their current use and/or promise in realizing the desired functional performance goals, as well as the quantity and quality of data available on human health impacts, life-cycle environmental impacts and material price (Supplementary Table 1).

ENM versus bulk chemical antimicrobial alternatives. To serve as a viable alternative to bulk chemicals or materials, ENMs should realize the same or enhanced functional performance while minimizing potential adverse impacts. Select chemical antimicrobials, including triclosan and triclocarban, were recently banned by the US Food and Drug Administration (FDA)[25]. This has increased interest in ENM-based antimicrobials, which have shown efficacy in water treatment and medical devices[27], but still require more robust environmental and human health impact assessments.

As a first step, designers must meet a specific functional performance goal when choosing substance alternatives. For example, for antimicrobial agents, the performance metric could be antimicrobial activity against a specific organism (for example, *Escherichia coli*) in terms of minimum inhibitory concentration (MIC, mg l^{-1}). Three implications are also evaluated to understand the trade-offs between the beneficial aspects of each material and its detrimental effects. The first is the price (in US$ kg^{-1}) that product designers pay to nanomaterial producers for the ENMs in their product. Nanomaterial price is a useful screening tool for nanomaterials selection because it is ubiquitously available and intercomparable. However, we note critically that price does not reflect the total cost to the user (that is, materials with lower starting material prices may translate to higher total cost to the manufacturer, due to various differences in the processing or other considerations, and this should be kept in mind when applying the selection strategy). The second implication metric is the CED (in MJ kg^{-1}) that comes from the production of nanomaterials. This is calculated from cradle-to-gate via life-cycle assessment, where the most sustainable material is the one that requires the least amount of energy during production. This includes electrical and heat energy during production, as well as the embodied energy of the precursors and reagents. The third consideration is hazard to human health. Life-cycle toxicity assessment does not account for physicochemical differences in nanomaterials, and does not always have the background data from experimental toxicity assays to support its assumptions[28], while toxicity assessment across multiple trophic levels of organisms is expensive and time-intensive. Therefore, in this study, relative human health hazard is indicated by the results of embryonic zebrafish toxicity assays (LC_{50}, the concentration of a material that is lethal to 50% of a population, mg l^{-1}), where the least toxic material is the one with the highest LC_{50} value. Since risk is a function of hazard and exposure[29], product designers can minimize risk by minimizing inherent hazard regardless of the circumstances of the exposure scenarios. Further, hazard is directly related to the physicochemical properties of a material, which can be tuned (see section 'Approach' in the Supplementary Information for more information), while exposure requires numerous assumptions, the breadth of which would be difficult to capture in an Ashby chart.

As expected, chemical and ENM performance, environmental/human health impacts and price exhibit a broad range (Fig. 2).

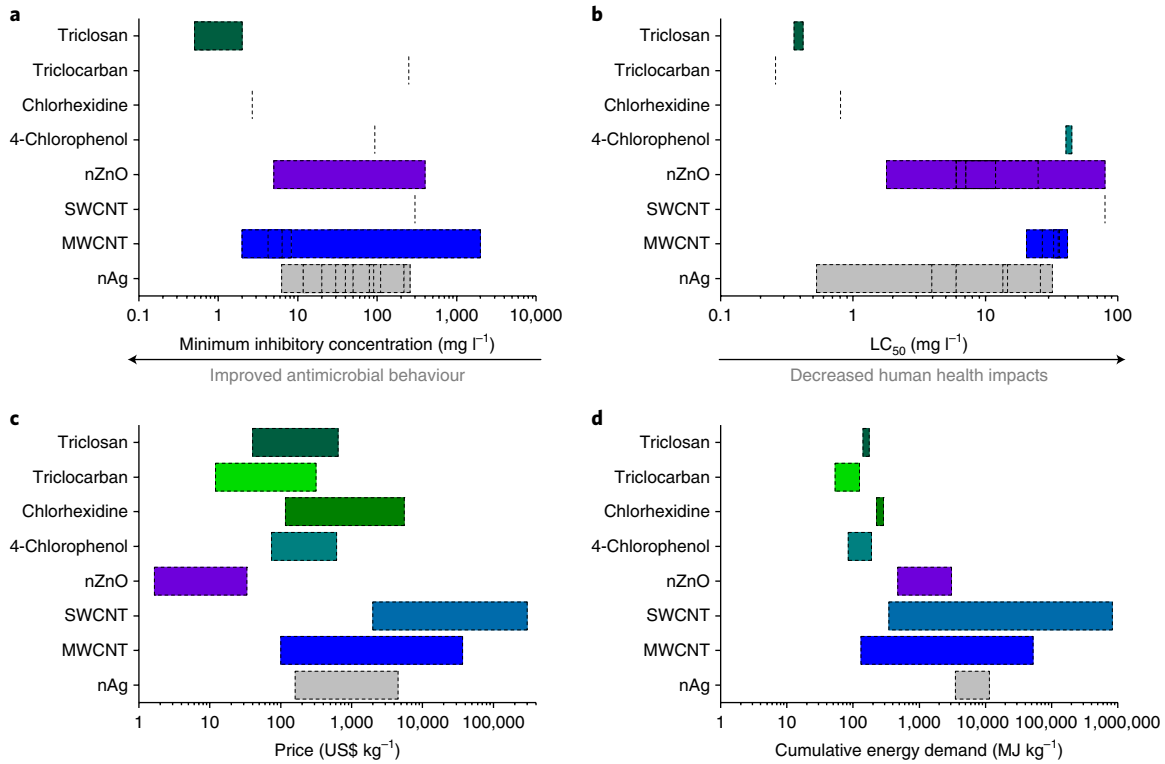


Fig. 2 | Performance of traditional antimicrobial chemicals and proposed nanomaterial substitutes. a–d, Efficacy (antimicrobial activity as MIC; **a**), environmental/human health impact (embryonic zebrafish toxicity as LC_{50}; **b**), price (**c**) and environmental impact (cumulative energy demand; **d**). Vertical dashed lines indicate single values taken or derived from literature sources and shaded bars encompass the entire range of values derived from literature sources.

Three of the four conventional chemicals considered pose greater human health concerns (lower LC_{50} value) than most of the ENMs, even when compared to the most concerning ENMs (for example, silver nanoparticles (nAg)). Chemical antimicrobials tend to require less energy in production and have a smaller range of energy requirements due to their commodity status, while many ENM synthesis methods are still being explored and developed. When considering antimicrobial efficacy and price, zinc oxide nanoparticles (nZnO) are economically competitive over some bulk chemical alternatives due to the combination of a relatively low MIC and unit price[30] (http://www.ebiochem.com/, https://www.tcichemicals.com/en/us/) (Fig. 3b).

When significant ranges exist within a material class, it suggests the opportunity to tune physiochemical properties to impart the desired outcome. As one example, multiwalled carbon nanotubes (MWCNTs) can either be one of the most promising or least effective antimicrobial agents, depending on the surface chemistry of the material[31,32] (Fig. 2a). Similarly, variability in ENM morphology (for example, size and shape) imparts a wide range in antimicrobial activity and embryonic zebrafish toxicity values, especially for nZnO and nAg, where the mechanism of action is at least partially related to ion production[33,34], whose dissolution rate is fundamentally related to surface area, strain or curvature, and crystallinity[35]. The wide ranges in CED values, especially for ENMs, are related to the large variety of synthesis routes for each material. These different synthesis routes are used to make ENMs of variable size or morphology, as is the case for many non-carbonaceous ENMs, or to synthesize materials of variable purity or structural quality, as is the case for carbonaceous ENMs[36]. The impact of these physicochemical differences and how they can be understood to inform material design is discussed further below.

When selecting a substance, one is often evaluating trade-offs or attempting a co- or multi-optimization scheme, and two-dimensional (2D) bubble charts (for example, relating the functional performance parameter to an environmental implication parameter) enable this multi-criteria comparison (Fig. 3). For example, a designer with established constraints of high efficacy and low hazard can quickly eliminate several options (that is, substances outside the upper left region in Fig. 3a), including triclosan and triclocarban, while still providing a variety of viable alternatives (for example, MWCNTs, nZnO, nAg and 4-chlorophenol). Given a basic constraint for performance or benefit, further specifications can be set for the maximum tolerable impact (for example, CED or material price; Fig. 3b,c) based on regulatory limitations, public pressure, industrial norms and the influence of other stakeholders. Similarly, multi-criterion thresholds can be set (for example, simultaneous optimization of performance, impact and price), and can be visualized in a multi-dimensional material parameter space (Fig. 3d). This concept can be naturally extended to n dimensions aided by a digital platform[37]. In this example, nZnO emerges as a viable substitute due to the low price, high LC_{50} (low hazard) and relatively low MIC (high efficacy).

Given the likelihood of having to balance multiple criteria, trade-offs can be delineated by selection guidelines. For example, selecting the least hazardous substance (highest LC_{50} value) with the highest efficacy (lowest MIC value) could be informed by maximizing the ratio of material functional performance to hazard, $R_{FH} = LC_{50}/MIC$. Accordingly, a selection guideline with a slope of R_{FH} can be plotted, where all substances along that line are 'equivalent' alternatives in terms of the ratio of their hazard and efficacy (Fig. 3a). Similar to the Ashby material selection index, M, these selection guidelines provide a mechanism for assessing a trade-off between the x and y axes; however, unlike M, R_{FH} does not have

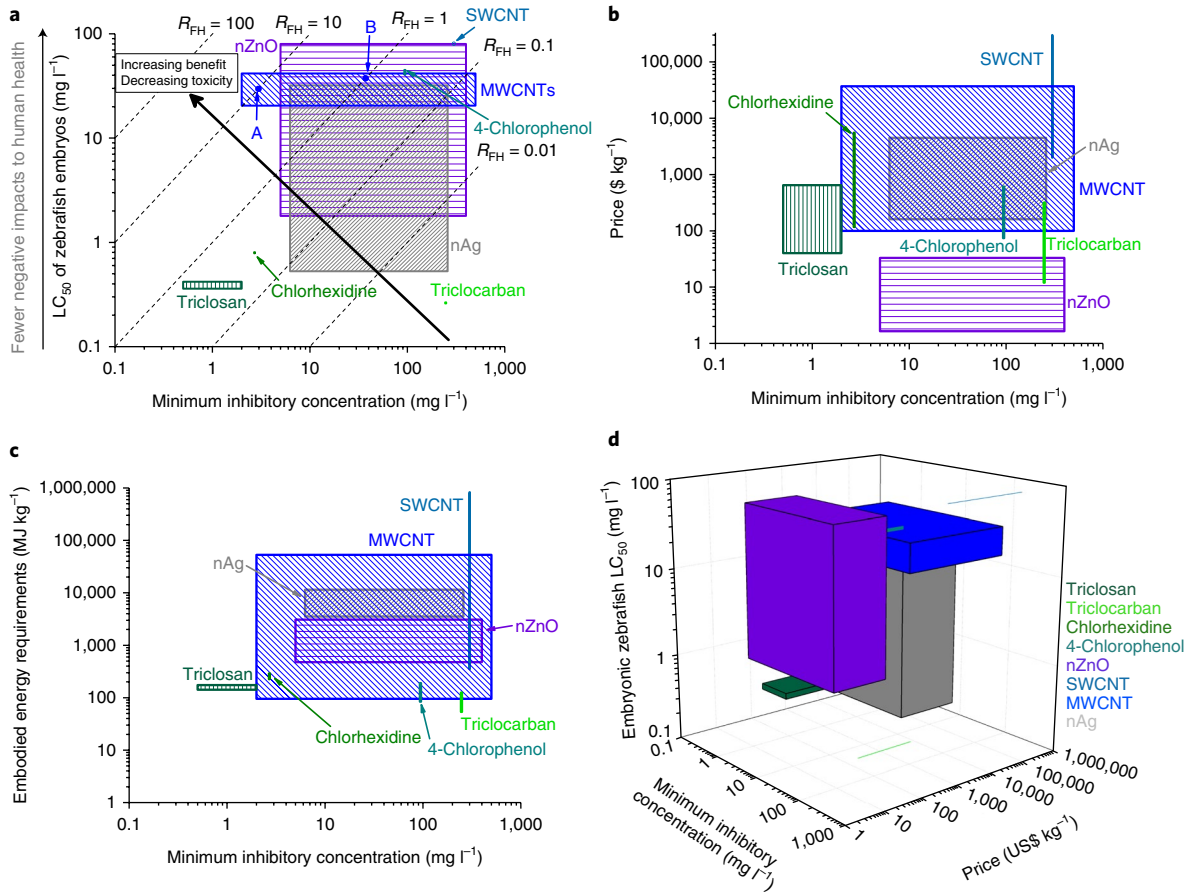


Fig. 3 | Co-optimization diagrams for chemical and nano-enabled antimicrobial agents. a–d, Functional performance (antimicrobial activity) compared to zebrafish toxicity (**a**), price (**b**) and cumulative energy demand (**c**), and 3D chart relating antimicrobial activity, embryonic zebrafish LC_{50} and price of nanomaterials (**d**). The dotted diagonal lines in **a** represent $R_{FH} = 0.01$ to $R_{FH} = 100$, where R_{FH} is the ratio of adverse impact to benefit, and points A and B represent different possible R_{FH} values for MWCNTs, which are dependent on the material's physicochemical properties.

a fundamental mathematical underpinning. These guidelines offer valuable information such that substances along the $R_{FH} = 10$ line have an LC_{50} that is ten times greater than those along the $R_{FH} = 1$ line for the same functional efficacy. Accordingly, the most desirable options for an antimicrobial agent (maximized R_{FH} value) are ENMs, including MWCNTs, nAg and nZnO.

There are several considerations to highlight: (1) for a single material class (for example, MWCNTs), R_{FH} can range across orders of magnitude due to differences in physiochemical properties within that class (see points A and B on Fig. 3a); (2) an $R_{FH} = 1$ does not necessarily indicate a net neutral ratio of a material's adverse impact to functional benefit, and instead should be considered as a relative metric to compare within or between material classes; (3) the definition of R_{FH} can vary from one material selection chart to another; and (4) the value of R_{FH} is a consequence of the metrics that are being compared, and as such, a value of R_{FH} for a particular selection diagram cannot be universally extended to other selection diagrams with different criteria on the *x* and *y* axes.

Comparing ENM-enabled composite materials for conductive polymers. Polymers are enabled with ENMs for many reasons, including to reduce ENM release, to provide stability from environmental transformation, or to enhance a functional performance property, such as the electrical conductivity of intrinsically conducting polymers such as PANI[38]. The ENM type and mass loading critically influence the net impact on conductivity, price and CED of the composite material (Fig. 4a,b and see Supplementary Fig. 1 for 1D visualization). Note that inherent material hazard is not considered here due to challenges in data availability, particularly as it relates to ENM release and exposure, and the lack of understanding regarding the toxicity of the composites themselves[39]. From the selection diagrams (Fig. 4), a constraint requiring a 10-fold higher conductivity for a PANI–ENM composite compared with neat PANI (Fig. 4b, constraint 1), and a stipulation that the CED of the additional ENM does not exceed 100 MJ kg^{-1} (Fig. 4b, constraint 2) can be set. These constraints narrow the available materials to single-walled carbon nanotubes (SWCNTs), MWCNTs and $nTiO_2$. Seeking to minimize the ENM price further narrows the choice to MWCNTs and $nTiO_2$, focusing innovation efforts to just two viable material candidates (see 3D co-optimization, Fig. 4c). In this way, performance, negative impact and material price can be considered simultaneously to facilitate sustainable product design.

Towards improved rational ENM design

Following initial materials screening and selection, fundamental relationships between ENM structure and physicochemical properties, functional performance and adverse impacts can help materials scientists tune desired outcomes[3]. For example, size, shape and surface chemistry are properties that can be partially or fully controlled a priori through synthesis and post-fabrication approaches, and all have been shown to significantly influence ENM function and hazard profiles[3,40,41]. Knowledge about these relationships can then provide tools to inform rational material design for improved functional performance and reduced adverse impacts.

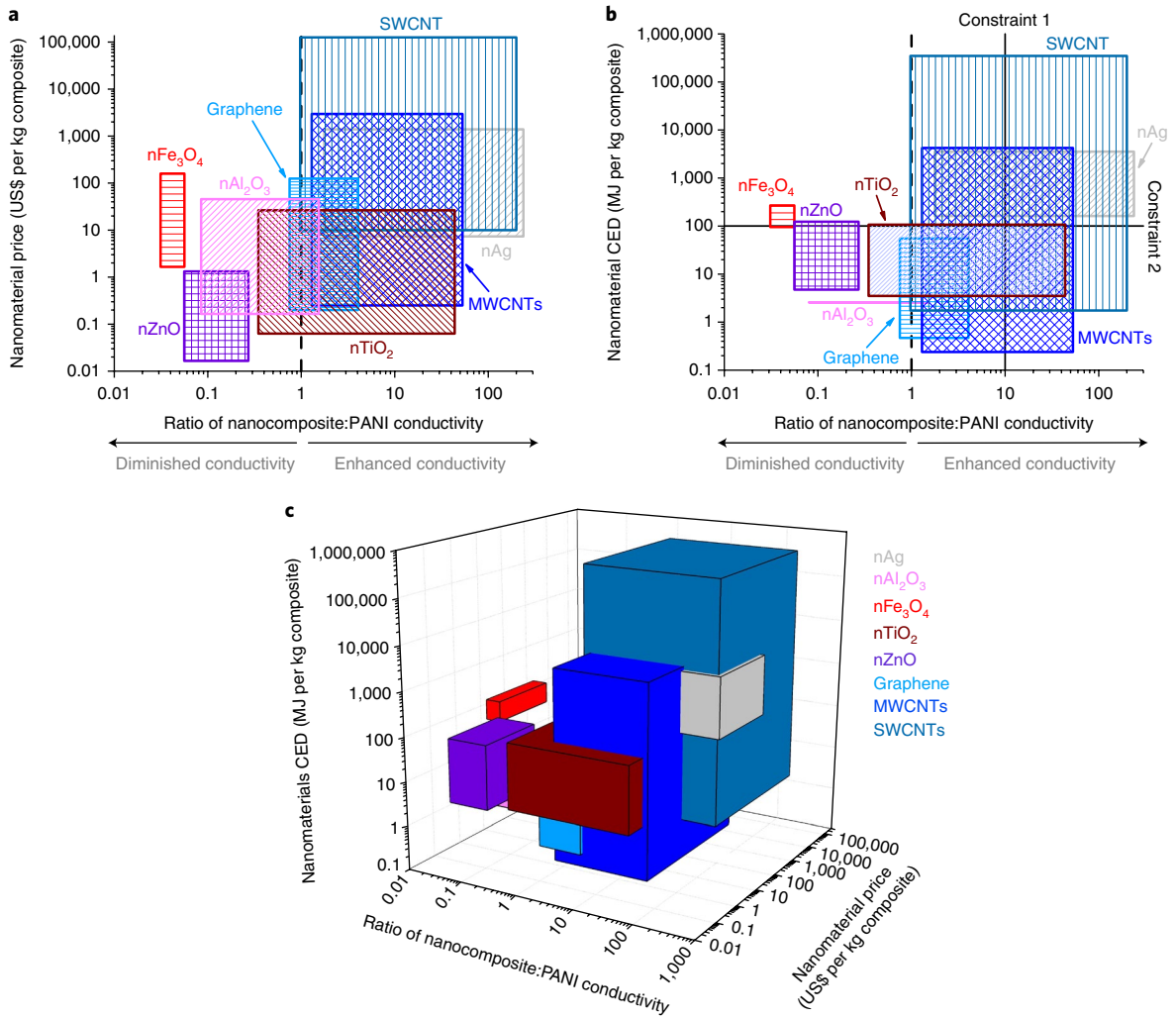


Fig. 4 | Selection charts for nano-enabled composites of polyaniline. a–c, Functional performance compared to the price of the nanomaterials in the composite (**a**) and cumulative energy demand (CED) of the nanomaterials in the composite (**b**), both weighted by proportional mass, and co-optimization diagram relating functional performance, price and CED (**c**). The dashed lines in **a** and **b** indicate where the nano-enabled composite has the same electrical conductivity as neat polyaniline, and ratios represent mass-based enrichments or depletions thereof. The solid lines in **b** indicate the hypothetical constraints that could be set by a product designer.

For example, nAg as an antimicrobial agent demonstrates that decreasing nanoparticle size resulted in higher antimicrobial activity (lower MIC), and higher relative lab-scale CED (Fig. 5a and Supplementary Tables 2–12). That is, the ENM with the highest efficacy (lowest MIC) has an environmental 'cost' (for example, high CED). This inherent trade-off highlights the need for improved syntheses for small nAg with lower CED to meet the desired efficacy constraint with more favourable environmental performance. The ENM–PANI case study reveals that nAl_2O_3 shape might influence composite conductivity and lab-scale CED (Fig. 5b and Supplementary Tables 13–16). Specifically, two nAl_2O_3 shapes enhanced PANI conductivity with drastically different CEDs; the nanoflake-nAl_2O_3–PANI has the highest conductivity with four-fold lower CED, as compared with the nanofibre. Thus, material information about functional and environmental performance can empower designers to enhance a product's functionality while minimizing adverse impacts.

Future considerations

While this strategy, like Ashby's, offers a promising way to select sustainable (nano)materials, all such approaches present some limitations[42]. Material selection charts, including this framework, are not meant to be deterministic tools to identify the optimal material for a specific application, but are instead meant to focus limited time and resources on a more promising and curated pool of material candidates. This pool itself may be constrained by the nature of the material considered; that is, material classes such as composites and alloys will not be comprehensively captured given the infinite possible material combinations. ENMs present further challenges to this approach in that small changes in any number of physiochemical structures or properties can significantly influence functional performance as well as hazard implications. This can result in a relatively large 'bubble' for a given nanomaterial class, thus confounding material selection due to overlap. As such, it is important that these approaches are part of a multiple step process in selecting appropriate (nano)materials by eliminating those that are not viable from consideration early on in the design process.

The expansion of material selection charts is essential for rational sustainable design of ENMs and ENM-enabled products. To date, safety assessments of ENMs are generally conducted in the absence of functional performance information, raising questions about the utility of the studies themselves (that is, the tested materials may

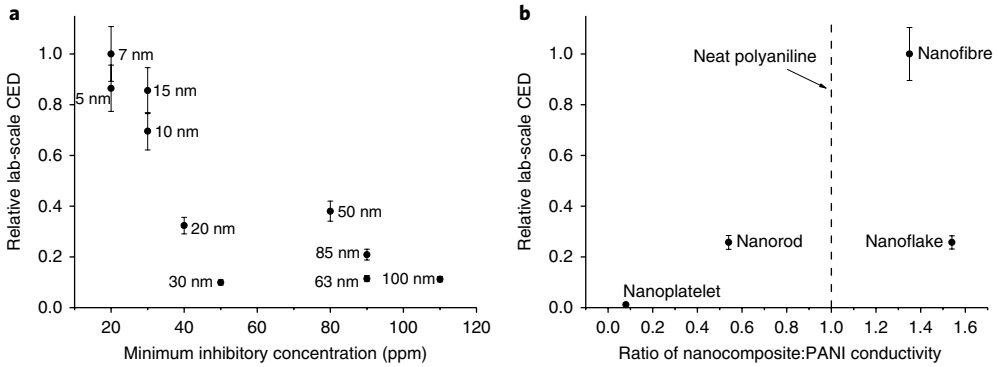


Fig. 5 | Nanomaterial functional performance and embodied energy as a function of physical properties (size and shape). a,**b**, Parametric relationship evaluation of nAg size on relative CED and antimicrobial activity (**a**) and influence of nAl_2O_3 shape on relative CED and PANI–ENM conductivity change (**b**), where the dotted line indicates a nanocomposite with the same electrical conductivity as neat polyaniline. CEDs shown here were calculated as part of this study, and error bars are the standard deviation of the CED values as a result of uncertainty analyses (see Supplementary Information).

not be most relevant or urgent from an application perspective). Likewise, designers cannot consider environmental impacts during development and commercialization because the appropriate data for their system may not be available. By defining a common set of data needs, limited nanoEHS research resources can focus on crucial data gaps to inform the development of design guidelines for sustainable ENMs and ENM-enabled products. In turn, materials scientists and engineers will have guidance toward the level of improvement they must achieve (via enhanced performance or mitigated implications) to be functionally competitive with the existing substances.

To realize these benefits, the relevant stakeholders must (1) agree on the type of data needed to efficiently construct material libraries and (2) systematize the data quality standards[43], so that all materials can be evaluated through the same or directly comparable process. Such standard practices exist for most traditional materials characterization approaches (for example, the ASTM methods for tensile strength[44]), but are lacking with respect to many of the novel applications of ENMs and for environmental impacts generally[45–47]. That is, disparate metrics or approaches reduce the synergistic power of the collective community to uncover parametric or fundamental relationships between ENM structure, properties and hazards[3]. The collection and inclusion of more data for consistent metrics using consensus protocols will improve chart utility for the general selection of chemical and material classes and will inform design. Similarly, criterion for environmental metrics beyond embodied energy, material recyclability and carbon footprint[21], such as ecotoxicity and cumulative water demand, are still under development. Further, improved consistency in ENM characterization and metrics can lead to the emergence of software-based platforms coupled to materials prediction from theory, which may ultimately include or enable calculation of hybridized and decorated materials. However, visualization of all of the possible materials space simultaneously may obfuscate the value of the Ashby strategy. Indeed, future users will likely have to specify what part of the materials space they wish to access and constrain whether or not they are including parameters calculated from theory or measured via experiment.

Conclusions

Leveraging conventional material selection frameworks that are focused on performance and material price, and which are already familiar to materials scientists and engineers, to consider ENMs and their environmental and human health impacts will be increasingly important as the ENM enterprise continues to develop and mature. The systematic selection strategy presented here offers multi-faceted practical benefits including (1) selecting the optimum substance considering a variety of attributes (for example, performance, environmental and human health implications, price of materials) from the currently available ENM, chemical and material libraries, (2) addressing real or perceived risks, and (3) avoiding regrettable substitution. This framework brings value to the material selection process, and supports curation of the ever-expanding nanomaterial library focusing limited resources on the most promising ENMs. Further, there are broadly transformative impacts of the proposed framework, such as highlighting data gaps to motivate critically needed research and developing tools to inform the design of ENMs and ENM-enabled products. This will contribute to the nanotechnology enterprise realizing its full potential to address societal challenges while doing so in a way that is both resilient, in the short term, and sustainable for future generations.

Data availability. The data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

Received: 18 July 2017; Accepted: 21 March 2018;
Published online: 30 April 2018

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Acknowledgements

This publication was developed under Assistance Agreement No. RD83558001 (LCNano) awarded by the US Environmental Protection Agency (EPA). It has not been formally reviewed by the EPA. The views expressed in this document are solely those of the authors, and the EPA does not endorse any products or commercial services mentioned in this publication. This work was also supported by the National Science Foundation (NSF) Nanosystems Engineering Research Center for Nanotechnology-Enabled Water Treatment (ERC-1449500; NEWT).

Author contributions

M.M.F., J.B.Z., D.L.P. and L.M.G. conceived and designed the reimagined framework. M.M.F. and S.S.C. performed the experiments and collected data. M.M.F. and S.S.C. analysed the data. M.M.F., S.S.C. and T.L.T. contributed materials/analysis tools. M.M.F., D.L.P., L.M.G. and J.B.Z. co-wrote the paper. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information is available for this paper at https://doi.org/10.1038/s41565-018-0120-4.

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